

Text Classification in NLP

Media Bias Detection, [Student 2 Topic], [Student 3 Topic]

Group 2 — CMPE346 Natural Language Processing

June 5, 2026

Group Overview

Shared setup

- ▶ Task: Text Classification
- ▶ Models: BiLSTM, BERT, RoBERTa
- ▶ Metric: Accuracy & Weighted F1

Individual tasks

- ▶ Atakan Gül — Media Bias Detection
- ▶ Student 2 — [Topic]
- ▶ Student 3 — [Topic]

Media Bias Detection in News

Task: Given a news sentence, classify it as *biased* or *not biased*.

Dataset: BABE (Bias Annotations By Experts)

- ▶ 4,121 news sentences from diverse outlets
- ▶ Expert-annotated binary labels
- ▶ 3,121 train / 1,000 test

Bias is expressed through framing and tone — not false facts. This makes it a genuinely hard NLP task.

Models

BiLSTM

Bidirectional recurrent network. Trained from scratch. Captures sequential patterns in both directions.

BERT

Transformer encoder pretrained on masked language modeling. Fine-tuned on BABE. Uses [CLS] token for classification.

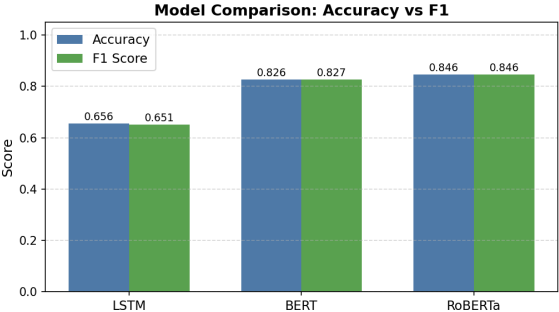
RoBERTa

Optimized BERT variant. Removes next sentence prediction, uses dynamic masking and more data. Consistently stronger on downstream tasks.

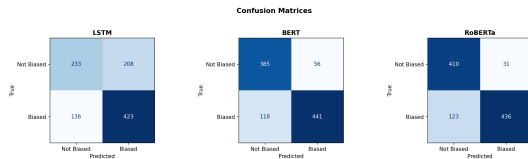
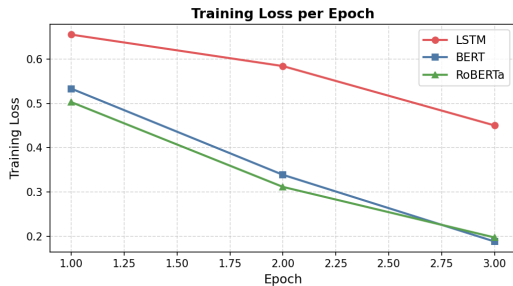
Results — Media Bias Detection

Model	Acc.	F1
BiLSTM	0.656	0.651
BERT	0.826	0.827
RoBERTa	0.846	0.846

RoBERTa best. +19% over LSTM.



Training Dynamics



Transformers converge faster and misclassify less. LSTM struggles on the biased class.

[Student 2 Topic]

Task: [Description]

Dataset: [Dataset name and brief stats]

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Student 2] — Key Finding

[Student 2 fills in: main insight from their results]

[Student 3 Topic]

Task: [Description]

Dataset: [Dataset name and brief stats]

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Student 3] — Key Finding

[Student 3 fills in: main insight from their results]

Conclusion

- ▶ Transformers consistently outperform BiLSTM across all three tasks
- ▶ RoBERTa is the strongest model in all experiments
- ▶ Pretraining is critical — LSTM trained from scratch cannot compete on small datasets
- ▶ Future work: domain-adaptive pretraining, larger model variants